



Undergraduate FYP Proposal Form¹

Section I: Student's Information:

Full Name	KHAW HAN ZHE		
Student ID	SWE2302061	Mobile	01131529001
Email ID	SWE2302061@xmu.edu.my	Programme	CST/SWE/DMT/CYS/DSC

Section II : Industry Information: (Complete this section if your project is going to be conducted in partnership with any company)

Company Name			
Contact Person			
Designation		Phone	
Email ID			

Section III : Declaration by the Student

By signing this form, I confirm that I have read and will adhere to the **Final Year Project Proposal Guidelines** of Xiamen University Malaysia as applicable to this application.

Signature of Student: _____

Date: 9/12/2025

Section VI : Supervisor Approval

Name of Supervisor	Hiew Fui Chin		
Office Location	A1#452		
Email Address	fuichin.hiew@xmu.edu.my		
Office Extension			
Signature		Date	10/12/2025

¹ The length of the proposal should be between 2000-3000 (Excluding proposed title & references)

SECTION V : Proposal Information

Proposed Title

Enhancing Indoor Plant Recommendations Through Solar Geometry and AR-Based Light Measurement

Introduction

A systematic review by Han et al. (2022) found that indoor plants have positive effects on the physiological and cognitive aspects of individuals, such as lowering blood pressure and improving academic performance. However, choosing suitable house plants that suit different light conditions of a house can be difficult for many non-expert indoor planters. Each house faces a different direction and daylight exposure, and the shading and illumination for each room depend on the sizes of the windows. All these conditions influence plant selection. Indoor planters normally rely on trial and error in choosing plants; therefore, failures in indoor planting are increasingly common Angel et al. (2025)

Many studies incorporate plant databases, environmental data collected from third-party API data sources, hardware sensors, or user input preferences, to recommend the most optimal choices of indoor plants. Several software systems were proposed, such as Green Oasis by Angel et al. (2025), which used a rule-based machine learning algorithm to suggest the best plant available for user-given information on room type, light availability, humidity levels, and a range of temperatures. However, these studies recommend plants and determine room lighting conditions based on fixed or static light labels (e.g. “low”, “medium”, or “bright”) from databases. **Most studies do not measure the distance, and at different times, the sunlight enters the window of a room. As a result, existing systems and studies cannot reliably recommend suitable plants for a specific location in a room.**

Solar geometry research has developed a solar positioning algorithm (SPA) to calculate solar zenith and solar azimuth angles from parameters, such as latitude, longitude, and time, with percentage errors up to $\pm 0.0003^\circ$. According to Reda & Andreas (2004), the accuracy the algorithm remains reliable from the year 2000 to 6000. In addition, multiple studies have shown that solar position computation can be incorporated into different fields to estimate illumination and irradiances in the industry of solar energy, as well as 3D modelling (Wang et al., 2026; Zhu et al., 2023; Liao et al., 2022). **However, the literature review found a lack of studies and utilization of solar geometry in indoor plant recommendation apps.** The integration of solar geometry and phone sensors may enhance the accuracy of indoor light information for indoor planters.

Therefore, this project proposes a mobile indoor plant recommendation app that utilizes the mobile phone sensor, solar geometry and Mobile Augmented Reality (Mobile AR) technology to acquire measurements of light and window sizes to determine the light condition. The mobile app will suggest suitable houseplants or the potential spots for plant placement based on the light conditions. This app may benefit ordinary and novice indoor plant lovers.

Aims and Objectives

The objective of this study is to design, implement, and evaluate an indoor plant recommendation app that assists novice indoor planters in choosing house plants for a specific location based on the light conditions. The recommendation app incorporates the mobile phone sensor, solar geometry and Mobile AR technology to measure light at a specific indoor location.

To achieve the aim of the project, four objectives are established:

- (i) To compare the accuracy and practicality of different Mobile AR distance and window measurement tools.
- (ii) To compare different plant recommendation techniques, such as rule-based or hybrid-based techniques, using common tropical indoor plant data, and benchmark the accuracy of the recommendation techniques.
- (iii) To design and implement the plant recommendation system, which incorporates mobile phone sensor, solar geometry and Mobile AR for light and spatial measurement.
- (iv) To evaluate the accuracy of the prototyped app in recommending indoor plants.

Literature Review

1. Light Measurement and Recommendation Techniques in Plant Recommender System

Most plant recommender systems studies aimed to help non-experts choose suitable plants for specific environments. These studies differ in terms of focus, data input methods, and algorithms. The review found that some studies do not provide actual sunlight measurements as input data, which may result in unreliable plant recommendations.

Doshi et al. (2019) designed a simple rule-based system that aims to enhance office air quality in the UAE based on volatile organic compounds, environmental temperature and user-specified plant size (small or large) for the office. However, the method that collects the required information inputs to the proposed system, such as sunlight conditions, was not mentioned in the paper. Angel et al. (2025) Green Oasis, which uses a rule-based algorithm to recommend plants based on room conditions, light availability, and plant care preferences. In this system, all required data to determine the plant recommendation is collected through manual user input. **The actual sunlight condition was not measured.**

Jaishree et al. (2025) propose an Analytic Hierarchy Process (AHP)-driven hybrid recommender for indoor and ornamental plants to solve a situation where the system failed to recommend products due to limited dataset availability in the system. However, the system relies on manually assigned ratings by experts as input for the sunlight requirements, such as “high”, “medium”, and “low”. Patil et al. (2025) introduce an AI-assisted plant recommendation system utilizing Random Forest Algorithm, manual user input and IoT sensors to collect plant growth factors, but not including sunlight, for plant suggestions. It only provides the sunlight requirement (e.g. “full sun”, “partial shade” and “semi-sun” and “shaded or indoor”) for each plant, which is predefined with other environmental profile values in the proposed system database. Likewise, Lodhi et al. (2023) proposed the

GroWiser, which incorporates IoT-based data collection and plant monitoring features (hardware components such as Arduino Uno and Node MCU, with temperature and humidity sensors in the plant pot). The study mentioned using an expert system model to recommend suitable plants, but the algorithm was not elaborated. **Similarly, actual sunlight was not measured in either study.**

Dhamecha (2021) included weather forecasting and geographical data into the proposed recommender system, with forecast-based input parameters such as climate zone, temperature, and humidity in a given area. Das et al. (2025) suggested a recommender system using a cosine similarity matching approach that ranks and suggests plants based on similarities between climate properties and plant species requirements. Both proposed solutions used ZIP codes and external API data as inputs to the recommender system. **However, the impact of their studies is mainly on outdoor agricultural plant selection.**

In essence, most proposed systems depend heavily on user-reported environmental conditions, especially lighting conditions, which **could introduce inaccuracies perpetually**. Sensor-based light measurement on real-time solar exposure is rarely mentioned in the reviewed papers, and none include indoor plant and placement suggestions in relation to windows and light variation measurement. In conclusion, existing implementations of plant recommender systems have yet to include AR spatial analysis for placement decisions.

2. The High Accuracy Capability of Solar Position Algorithm (SPA) and Its Applications

Reda & Andreas (2004) proposed and formalised the Solar Position Algorithm (SPA) as a method that offers a high accuracy of solar zenith and azimuth computing. The algorithm was mainly for solar radiation-related applications. Their work and studies are highly cited in any studies that involve the calculation of sun position. Many authors have incorporated SPA or related solar geometry routines in different domain-specific systems. For example, Guan et al. (2021) has developed a prototype system that evaluates the freeboard of icebergs in various sizes and volumes using the height of the icebergs' shadow, and the iceberg freeboard are retrieved from bi-temporal Landsat-8 panchromatic images. The system depends on accurate solar elevation values derived from SPA calculation to evaluate iceberg height from the height of the iceberg shadow.

Liao et al. (2022) incorporated SPA indirectly through an open source library, Pysolar, to compute hourly solar irradiation at 22 weather and solar monitoring stations using a combination of atmospheric data from satellite and precise irradiation calculation. The study has accurately estimated land-surface solar irradiation over a large region across Australia and China.

H. Wang et al. (2026) targeted Hong Kong's dense urban environment, where they have developed a framework that integrates LiDAR technology in 3D modelling with other ray tracking algorithms and solar radiation and solar positioning data from satellite, which are derived from SPA. The data obtained is used as parameters for their ray-tracing model to estimate how much solar energy can be collected and utilized for all rooftops of the buildings in a complex city environment.

Zhu et al. (2023) introduced an approach that focuses on outdoor illumination using AR technology with the assistance of SPA and other geospatial data sources, such as Landsat imagery data and atmospheric clarity indicator data, as inputs to shadows and lighting that mirror outdoor lighting at any moment. All the frameworks, including the utilization of SPA, are to make sure virtual objects generated in AR are consistently lit in corresponding with the given environment.

Dey et al. (2025) used Sentinel-2 satellite data to acquire the leaf area index (LAI) of agricultural crops. Their study has compared heavily how multiple modelling strategies can provide an evaluation of the accuracy of plant canopy density inferred from multispectral imagery. Solar geometry derived from SPA was used to support accurate canopy reflectance-related studies in the paper.

In summary, the proposed works have demonstrated that SPA possesses great capability in satellite imagery registration and interpretation, which can be used to predict geometric heights from object shadows. It was applied for the assessment of potential solar energy utilization in an urban environment and driving photorealistic lighting in AR technology advancement. **However, there is a lack of utilization of SPA for estimating light conditions in an indoor environment, and there is no existing study that incorporates SPA as part of the existing proposed indoor plant recommendation system.** In this project, SPA would similarly serve as the backbone of the plants recommender system that predicts sun path relative to window sizing and predicts sunlight duration to assist in recommending suitable plants for users.

3. Augmented Reality (AR) Technology in The Plant-related Domain

Multiple studies have explored Augmented Reality (AR) as a way to visualize or preview indoor plant placement. PlantAR proposed by Wang & Chang (2024) scans and overlays suggested plants with AR and AI level of precision. But the paper does not reveal much information on how environmental data, including light and humidity, are collected to determine location suitability for plant placement.

Shinde et al. (2024) proposed an ARCore-based mobile and web solution that allows users to preview plants at home and buy them online. The proposed system utilizes plane identification and Concurrent Odometry and Mapping to offer high 3D stability. The paper focuses on plant placement but not on a plant recommendation system.

ARPlant developed by Ibrahim et al. (2022) is an AR Malaysia rainforest plant learning app using AR markers technology with the Mobile Application Development Lifecycle Model (MADLC) process. The authors created 3D models and tested the usability with the System Usability Scale, with an excellent mean SUS of 87.81. But the system is entirely educational and is not able to estimate light or suggest plants depending on circumstances.

Afzal et al. (2024) created PlantView, which is a combination of a large 48-species CNN classifier with Unity/AR Foundation and can identify plants and superimpose 3D models and textual data. It is well developed in terms of species recognition and AR rendering.

In conclusion, studies report that AR is useful in visualising and recognising plants. **However, no studies integrate solar geometry and AR spatial measurement to estimate light suitability for recommending suitable plants or a specific location for plant placement, which leaves gaps that this study seeks to fill.**

Research Methodology

This study implements a prototype-based design and evaluation methodology where the system will be designed and implemented iteratively. The proposed system will be evaluated on its reliability in suggesting suitable plants and locations for plant placement.

1. System design, tools, and technologies

The system will be divided into three modules:

- (i) Solar geometry and light suitability module
- (ii) AR-assisted mobile window measurement and spatial placement module
- (iii) Mobile-based plant recommendation interface

Figure 1 illustrates the System Architecture of the Proposed Plant Recommendation System and data flow between the system modules, from SPA computation and mobile sensor inputs to AR plant placement visualization.

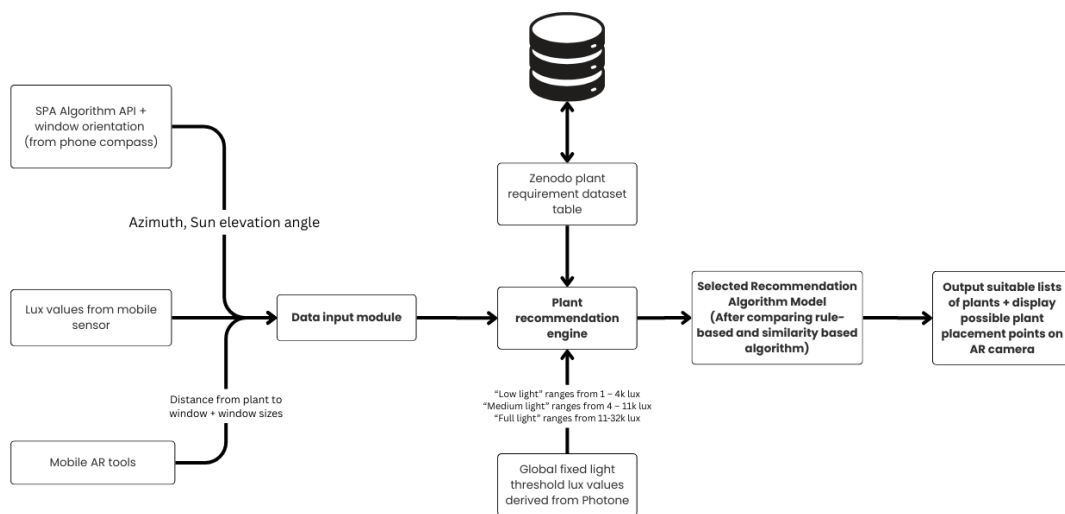


Figure 1
System Architecture of the Proposed Plant Recommendation System

The sun position at user's given location will be computed using Solar Position Algorithm (SPA) via an API relevant to the NREL SPA algorithm.

Android-specific AR frameworks will be used to measure window size and window plane, and the distance from the window to the plant spot using the phone camera.

The mobile sensor will be used to collect illuminance at the user's position and check for any potential shading outside the windows.

A backend web service and relational database will store the plant and measurement dataset. The front-end application will display suitable plant choices and placement locations using AR.

2. Recommendation Algorithm and AR spatial measurement comparison

- (i) Comparison of AR tools for spatial measurement: The most suitable AR toolkit will be chosen for the implementation plant-window distance and window-measurement module after comparing the measurement accuracy among them.
- (ii) Recommender algorithm comparison and evaluation: The recommended plant lists suggested by the content-based and simple rule-based algorithms will be compared against expert ratings or an independently constructed plants dataset. Both algorithms will be evaluated using precision, recall, and hit-rate metrics.

3. Data collection and datasets

Three important datasets will be included in the study:

- (i) Plant requirement and light lux threshold dataset

Plant requirements will be compiled from available indoor plant datasets. Ranged threshold lux values will be assigned to light categories (low, medium, and full light) from the plant dataset to determine suitable plants for the current lighting conditions in the room.

The plant requirements dataset will be obtained from the *Indoor plants DataSet* on Zenodo repository (Bouhadi et al., 2022).²

- (ii) Solar geometry and light suitability dataset

For a specific spot in a room, the recommendation system will record:

- (a) Solar geometry API data (sun azimuth, elevation angle, estimated sunlight duration, and window orientation) using smartphone compass.
- (b) Light brightness level (lux) using smartphone light sensor at different time of the day.

- (iii) AR-based window sizes and spatial measurement dataset

Android AR module will be used to estimate approximate window sizings (window width, height and sill height, and the distance between the plant placement point and the window).

² Bouhadi, M., Deroussi, A., Taji, K., & Morchid, A. (2022). *Indoor plants DataSet* [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.6984514>

4. Evaluation methods

The evaluation of the system will be conducted in one of two ways:

- (i) Accuracy of AR technology and sensor measurement: AR-based window and plant-window distance will be compared against actual tape measurements.

Mobile sensor-collected illuminance and SPA light-class prediction from API will be compared with lux meter for misclassification rates summarized using descriptive statistics.
- (ii) Reliability of recommendation system: Two strategies to test the reliability of the recommendation system are planned.
 - a. Final output plants list from the recommendation system will be compared against the plants' data from the ground-truth table constructed from multiple independent horticultural sources with precision, recall, and hit metrics to evaluate the reliability of the system.
 - b. Experienced indoor gardeners or horticultural experts will evaluate and rate the recommended plants from the system with testing scenarios given through questionnaires.

Scope & Limitations

The indoor urban plant recommendation system is implemented with a scope limited on tropical plants. Due to resource constraints, the prototype will be tested in Malaysia only with plant species that are common across tropical countries.

The software developed is limited to Android smartphones that support AR with basic phone sensors. This study focuses on the indoor urban residential environments; greenhouses and outdoor environments are excluded. The samples of homes and windows are limited to Malaysian urban apartments, condominiums and terrace houses with commonly seen sliding or casement windows.

Since there is no publicly available pre-trained plant recommendation model, a simple rule-based or similarity-based model will be compared and employed to implement the plant recommendation system.

Potential Project Significance

The project may contribute a functional plant recommender to the urban region residents, specifically novice indoor plant users. The plant recommender systems have the potential to address gaps in the reviewed studies. By incorporating solar position algorithms, phone light sensors, and AR-based spatial measurement technology, this study may provide a more objective way to identify light suitability for a specific spot for plant placement. This may help novice planters select suitable plants and plant placement, and enhance their capability in indoor plant care. Increasing the success of planting allows planters to benefit from the

physiological and cognitive aspects, such as lowering blood pressure and improving cognitive performance.

In addition, the datasets on plants required light conditions, the findings on using AR spatial measurements and recommender algorithm comparisons, can potentially serve as useful references for future work and as foundation for perfecting the indoor plant care and monitoring system.

Expected Outcomes & Concluding Remarks

This project is expected to produce a working plants recommender prototype with SPA-based sun-path prediction, AR window and distance measurement, mobile light-sensor data collection and a dataset for popular tropical indoor plants. Furthermore, it is expected that a dataset of tropical plants data could be used by other researchers.

In conclusion, it is expected that solar geometry demonstrates its capabilities to serve the field of indoor plant care and benefit plant recommendation systems. The prototype app will can measure light conditions at a specific location and recommend appropriate indoor plants with an optimal spot for placing the recommended plants.

Key References

- Afzal, S., Khan, H. A., & Lee, J. W. (2024). PlantView: Integrating deep learning with 3D modeling for indoor plant augmentation. *Ecological Informatics*, 84, 102899. <https://doi.org/10.1016/j.ecoinf.2024.102899>
- Angel, Y., Reddy, P. A. K., . Kv., Reddy, B. L., Kumar, B. K., & Kumar, M. P. (2025). Green oasis: A personalized indoor plant recommendation engine. *International Journal of Agriculture Extension and Social Development*, 8(4), 605–608. <https://doi.org/10.33545/26180723.2025.v8.i4i.1840>
- Das, S., Patra, A., Basu, A., & Nath, S. (2025). A Cosine Similarity-Based Framework for Plant Recommendation Using Indian Agro-Climate and Soil Data. *Volume 1*, 21–29.
- Dey, S., Saha, K., Dave, R., P, N., & Murugesan, A. (2025). Estimation of leaf area index of mustard and potato from Sentinel-2 data using parametric, non-parametric and physical retrieval models. *Remote Sensing Applications: Society and Environment*, 37, 101493. <https://doi.org/10.1016/j.rsase.2025.101493>
- Dhamecha, M. (2021). Plant Recommendation System with the Use of Weather Forecasting. *CEUR Workshop Proceedings*, 3058.
- Doshi, G. J., Stevens, J. D., & Khan, S. T. (2019). A Rule Based Expert System to Advise on Air-Filtering Plants for Indoor Spaces in UAE. *Proceedings of the 2019 5th International Conference on Systems, Control and Communications*, 82–86. <https://doi.org/10.1145/3377458.3377470>

- Guan, Z., Cheng, X., Liu, Y., Li, T., Zhang, B., & Yu, Z. (2021). Effectively Extracting Iceberg Freeboard Using Bi-Temporal Landsat-8 Panchromatic Image Shadows. *Remote Sensing*, 13(3), 430. <https://doi.org/10.3390/rs13030430>
- Han, K.-T., Ruan, L.-W., & Liao, L.-S. (2022). Effects of Indoor Plants on Human Functions: A Systematic Review with Meta-Analyses. *International Journal of Environmental Research and Public Health*, 19(12), 7454. <https://doi.org/10.3390/ijerph19127454>
- Ibrahim, N., Osman, R., Ali, A., Abdul Hamid, N., Nordin, S., Shahimi, S., Aminuddin, M. S., & Abdul Razak, F. (2022). Integrating Augmented Reality in Learning Plants. *IOP Conference Series: Earth and Environmental Science*, 1019, 12051. <https://doi.org/10.1088/1755-1315/1019/1/012051>
- Jaishree, Manish Madhava Tripathi, & Anshul Mishra. (2025). Hybrid recommender system for indoor and ornamental plants: An AHP driven approach. *International Journal of Applied Mathematics*, 38(6s), 331–355. <https://doi.org/10.12732/ijam.v38i6s.402>
- Liao, X., Zhu, R., & Wong, M. S. (2022). Simplified estimation modeling of land surface solar irradiation: A comparative study in Australia and China. *Sustainable Energy Technologies and Assessments*, 52, 102323. <https://doi.org/10.1016/j.seta.2022.102323>
- Lodhi, T., Qamar, T., Hussain, A. N., Saleem, H., & Khan, S. (2023). GroWiser: A Plant Recommendation and Monitoring System. *International Conference On Biological Research And Applied Science*. <https://doi.org/10.37962/IBRAS/2023/77-80>
- Patil, U. G., Gavali, P., Salmote, N., Chopade, M., & Bonde, H. (2025). Artificial Intelligence Based Domestic Plant Selection for Optimum Sustainability. *Metallurgical and Materials Engineering*, 31(4), 993–1001. <https://doi.org/10.63278/1543>
- Reda, I., & Andreas, A. (2004). Solar position algorithm for solar radiation applications. *Solar Energy*, 76(5), 577–589. <https://doi.org/10.1016/j.solener.2003.12.003>
- Shinde, R., Raut, A., Arjun, A., Bankar, H., & Dhage, S. (2024). Location Based Plant Selection through Augmented Reality. *International Research Journal of Engineering and Technology*. www.irjet.net
- Wang, H., Chen, B., & Gong, P. (2026). High-resolution 3D solar potential assessment reveals significant shading-driven overestimation in dense urban areas. *Landscape and Urban Planning*, 265, 105521. <https://doi.org/10.1016/j.landurbplan.2025.105521>
- Wang, Y., & Chang, M. (2024). Enhancing Indoor Environments through Augmented Reality and Artificial Intelligence for Personalized Plant Integration. *CS & IT Conference Proceedings*, 14(17).

Zhu, K., Liu, S., Sun, W., Yuan, Y., & Wu, Y. (2023). A Lighting Consistency Technique for Outdoor Augmented Reality Systems Based on Multi-Source Geo-Information. *ISPRS International Journal of Geo-Information*, 12(8), 324. <https://doi.org/10.3390/ijgi12080324>

Gantt Chart

No.	TASK2	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
1	Finalise requirements and with use cases with supervisor and set up project														
2	Detailed design of modules (SPA engine, AR, sensor, recommender, DB schema														
3	Outlining plant dataset curation (species, light needs, attributes)														
4	Implement SPA module and compass integration														
5	Compare ,evaluate and implement AR techonology in window plane and dsitance measurement prototype														
6	Implement light sensor for illuminance data collection & manual user input														
7	Algorithm comparison and evaluation														
8	Recommendation models implementation														
9	Mobile UI and Integration														
10	Dataset accuracy evaluation														
11	Final deliverable system testing														
12	Documentation, thesis and analysis writings and discussion														

² Please follow APA style of referencing.